# Information Spread on Online Networks

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# Background Information

- Information spread refers to how quickly a piece of information gets distributed online and also how many people it reaches
- There are many different factors that influence this
- One of the most common ways to depict this phenomenon is the basic SIR model
- This principal can be expanded to more complex models

# Questions to Consider

- In the long term, what proportion of the population is informed?
- What is the proportion of the population that knows the information after 24 hours?
- How do these values compare between the different models?

# Basic UID Model

- Similar to the basic SIR model, except titles of classes are changed
- Equations:

$$\frac{dU}{dt} = -\alpha UI$$

$$\frac{dI}{dt} = \alpha UI - \beta I$$

$$\frac{dD}{dt} = \beta I$$

- The parameter $\alpha$ represents the transmission coefficient and $\beta$ represents the recovery rate

# Model Development and Assumptions: Modified UID Model

- The UID model is expanded to include a reinfection rate
- Equations:

$$\frac{dU}{dt} = -\alpha UI$$

$$\frac{dI}{dt} = \alpha UI + \gamma ID - \beta I$$

$$\frac{dD}{dt} = -\gamma ID + \beta I$$

- The parameters $\alpha$ and $\beta$ are the same as the previous model, but $\gamma$ represents the reinfection rate

# Analysis and Results: Modified UID Model

- The modified UID model can be reduced to two equations so simplify analysis
- The population size is assumed to be constant, so we can make the substitution N = U +I+ D

$$\frac{dU}{dt} = -\alpha UI \qquad \frac{dI}{dt} = (\alpha - \gamma)UI + I(\gamma N - \gamma I - \beta)$$

- To find the proportion of the population in the Disinterested class, we can substitute to find it later

# Analysis and Results: Modified UID Model

- Then the equilibria are: $(0,0),\ \left(0,\ N - \frac{\beta}{\gamma}\right),\ \left(\frac{\gamma N - \beta}{\gamma - \alpha}, 0\right)$

- The second equilibria is the case where the information spread persists, which is the most interesting case, so this is where analysis will be focused
  - $(0,0)$ is the trivial equilibrium
  - $\left(\frac{\gamma N - \beta}{\gamma - \alpha}, 0\right)$ is the case where the information spread dies out

- Restrictions for the second equilibria are $N - \frac{\beta}{\gamma} > 0$ or $\frac{N\gamma}{\beta} > 1$

- This value is known as the basic reproduction number, $R_0$

# Analysis and Results: Modified UID Model

- Stability of the equilibria can be determined using the Jacobian matrix

$$J = \begin{bmatrix} -\alpha I & -\alpha U \\ \alpha I - \gamma I & \alpha U + \gamma(N - U - I) - 2\gamma I - \beta \end{bmatrix}$$

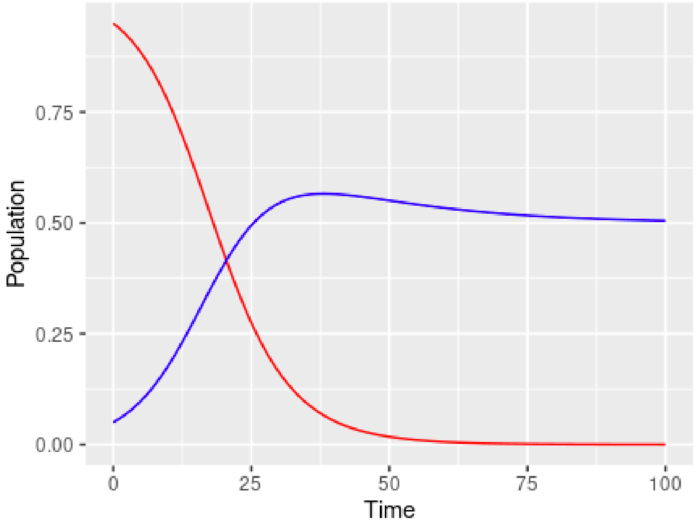
$$J\left(0,\, N - \frac{\beta}{\gamma}\right) = \begin{bmatrix} \frac{-\alpha}{\gamma}(\gamma N - \beta) & 0 \\ \frac{(\gamma N - \beta)(\alpha - \gamma)}{\gamma} & -2(\gamma N - \beta) \end{bmatrix}$$

- The eigenvalues are $\frac{-\alpha}{\gamma}(\gamma N - \beta)$ and $-2(\gamma N - \beta)$
- Both of these values are negative, which means that the equilibrium is a sink

# Analysis and Results: Modified UID Model

- After running simulations for the model, we can analyze the long term behavior
- Parameters/Initial Conditions:

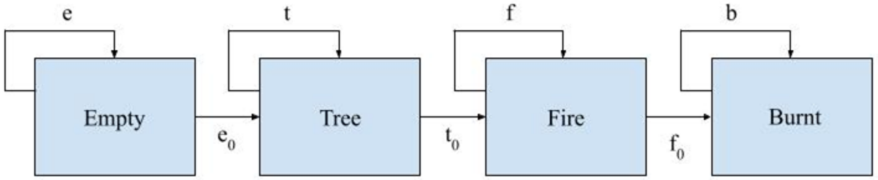
| | |
|---|---|
| $\alpha = 0.2$ | $N = 1$ |
| $\beta = 0.05$ | $U_0 = 0.95$ |
| $\gamma = 0.1$ | $I_0 = 0.05$ |




- The population stabilizes with no one in the Unaware class, 50% of the population in the Informed class, and 50% of the population in the Disinterested class

# Model Development and Assumptions: Matrix Model

- The algorithm in the modified forest-fire model is adapted into an absorbing Markov chain model
- The population is divided into four classes: Empty, Tree, Fire, and Burnt




- The Burnt state is an absorption state in the model

# Analysis and Results: Matrix Model

- Based off of the compartmental diagram, the transition matrix can be formulated

$$T = \begin{bmatrix} e & 0 & 0 & 0 \\ e_0 & t & 0 & 0 \\ 0 & t_0 & f & 0 \\ 0 & 0 & f_0 & 1 \end{bmatrix} \Rightarrow A = \begin{bmatrix} e & 0 & 0 \\ e_0 & t & 0 \\ 0 & t_0 & f \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & f_0 \end{bmatrix}, O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, I = \begin{bmatrix} 1 \end{bmatrix}$$

- The elements of this matrix are important for finding the steady-state matrix, L

$$L = \begin{bmatrix} O_{3x3} & O_{3x1} \\ B(I-A)^{-1} & I_{1x1} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{-e_0 f_0 t_0}{(e-1)(f-1)(t-1)} & \frac{f_0 t_0}{(f-1)(t-1)} & \frac{f_0}{1-f} & 1 \end{bmatrix}$$

# Analysis and Results: Matrix Model

- The fundamental matrix, F, is shown below

$$F = (I - A)^{-1} = \begin{bmatrix} \frac{1}{1-e} & 0 & 0 \\ \frac{e_0}{(e-1)(t-1)} & \frac{1}{1-t} & 0 \\ \frac{-e_0 t_0}{(e-1)(f-1)(t-1)} & \frac{t_0}{(f-1)(t-1)} & \frac{1}{1-f} \end{bmatrix}$$

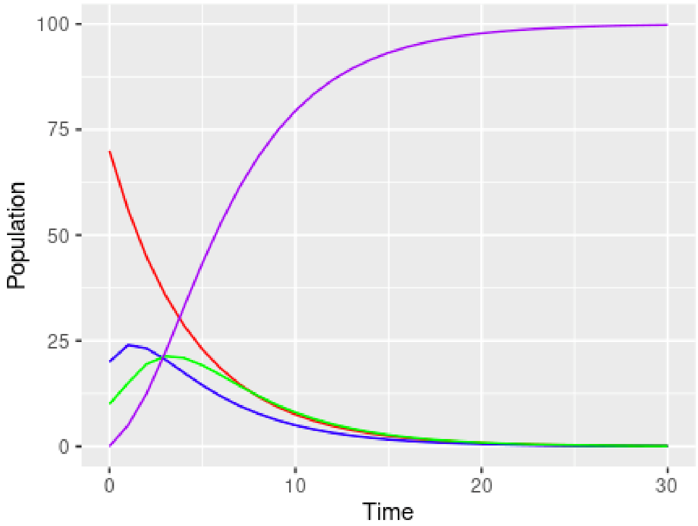
- After substituting the specified parameters, the matrices L and F become

$$L = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \qquad F = \begin{bmatrix} 5 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$$

# Analysis and Results: Matrix Model

- Similar to the previous model, we can analyze the long-term behavior through simulations
- Parameters/Initial Conditions:

| | |
|---|---|
| $e = 0.8$ | $E_0 = 70$ |
| $e_o = 0.2$ | $T_0 = 20$ |
| $t = 0.5$ | $F_0 = 10$ |
| $t_o = 0.5$ | $B_0 = 0$ |
| $f = 0.5$ | |
| $f_o = 0.5$ | |




- Since this model includes an absorption state, everything will end up there in the long term

# Model Comparison

- After 24 hours, the modified UID models predicts:

  30.3% in Unaware
  47.9% in Informed
  21.8% in Disinterested

- After 24 hours, the matrix model predicts:

  0.47% chance of Empty
  0.31% chance of Tree
  0.52% chance of Fire
  98.69% chance of Burnt

# Conclusions and Future Work

- Moving forward, there are some adjustments that can be made to the models
- Parameter adjustment to become more consistent between models
- A comparison of the Informed class and Fire state because both represent the information actively spreading